

AI Research Scientist V2

Multi-Agent AI Research Automation System

1. Problem Analysis

Business Context

Research and innovation require extensive effort in collecting existing knowledge, reviewing literature, identifying research gaps, designing experiments, and preparing scientific documentation.

Researchers and students spend significant time performing repetitive tasks such as literature analysis, organizing research ideas, generating citations, and formatting reports.

The AI Research Scientist V2 system addresses this challenge by automating the research workflow using multiple specialized AI agents. Each agent handles a specific research task while maintaining shared context throughout the process.

Stakeholders

Stakeholder	Requirements
Researchers	Faster literature review and research assistance
Students	Support for academic research and report generation
Organizations	Automated knowledge discovery and analysis
Domain Experts	Review and validate AI-generated outputs
Developers	Demonstrate scalable multi-agent AI systems

Problem Statement

Develop a multi-agent AI research assistant capable of automating the research lifecycle, including research planning, literature review, gap identification, experiment design, citation generation, and scientific report creation.

Objectives

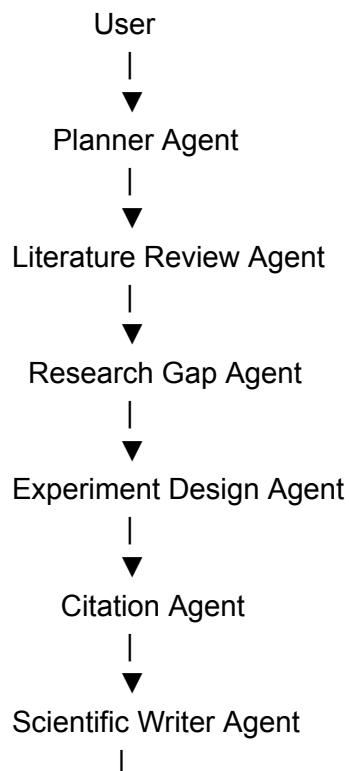
The objectives of the system are:

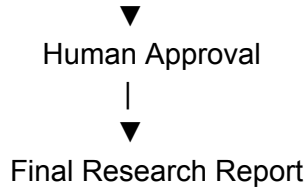
- Automate repetitive research activities using AI agents
- Divide complex research workflows into specialized tasks
- Enable collaboration between multiple AI agents
- Maintain shared context across agent interactions
- Generate structured research outputs
- Provide human approval before final report generation

2. Multi-Agent Design

Agent Architecture

The system follows a sequential multi-agent architecture where specialized agents execute different stages of the research workflow.





Roles of Each Agent

1. Planner Agent

Purpose

Creates a structured research roadmap based on the given topic.

Responsibilities

- Understand the research objective
- Break down the topic into research tasks
- Define research direction
- Create an execution plan

2. Literature Review Agent

Purpose

Analyzes existing research and summarizes current knowledge.

Responsibilities

- Identify important research concepts
- Summarize existing approaches
- Analyze current trends
- Extract important findings

3. Research Gap Agent

Purpose

Identifies limitations and unexplored areas in existing research.

Responsibilities

- Analyze literature findings
- Identify unresolved problems
- Suggest future research directions

4. Experiment Design Agent

Purpose

Designs a possible research methodology.

Responsibilities

- Suggest datasets
- Define experimental approaches
- Recommend evaluation metrics
- Predict expected outcomes

5. Citation Agent

Purpose

Generates academic references for the research report.

Responsibilities

- Generate APA-style citations
- Organize references
- Support scientific documentation

6. Scientific Writer Agent

Purpose

Creates the final research report.

Responsibilities

- Combine outputs from all agents
- Structure research findings
- Generate complete Markdown report

Agent Interaction and Handoff Flow

Research Topic



Planner Agent



Research Plan



Literature Review Agent



Literature Findings



Research Gap Agent



Identified Research Gaps



Experiment Design Agent



Research Methodology



Citation Agent



Scientific Writer Agent



Human Approval



Final Research Report

Tool Integration Overview

GitHub Models API

Purpose:

- Provides LLM capabilities
- Enables reasoning and content generation
- Powers all specialized AI agents

OpenAI Python SDK

Purpose:

- Handles communication with the language model API
- Provides standardized API interaction

Python Runtime

Purpose:

- Executes agent workflows
- Handles state management
- Controls agent execution order

Rich Library

Purpose:

- Provides terminal interface enhancements
- Displays workflow progress

python-dotenv

Purpose:

- Manages environment variables
- Secures API configuration

3. Implementation

Specialized AI Agents Implemented

The system implements six specialized AI agents:

Agent	Responsibility
Planner Agent	Research planning
Literature Review Agent	Existing research analysis
Research Gap Agent	Gap identification
Experiment Design Agent	Research methodology design
Citation Agent	Reference generation
Scientific Writer Agent	Final report generation

Agent Handoffs

Each agent receives the output generated by the previous agent through a shared research state.

Workflow:

Agent Output



Shared Research State



Next Agent Input

This allows agents to collaboratively build a complete research report while maintaining context.

Memory / Context Management

The system maintains a centralized research state that stores information generated throughout the workflow.

Stored information includes:

- Research topic
- Research plan
- Literature analysis
- Research gaps
- Experiment methodology
- Citations
- Final report content

The shared state enables seamless information transfer between agents.

Structured Outputs

Each agent produces structured Markdown outputs stored separately.

Generated artifacts:

reports/

- |— plan.md
- |— literature.md
- |— gaps.md
- |— experiment.md
- |— citations.md
- |— final_report.md

This modular structure allows individual agent outputs to be reviewed and reused.

Human Approval

The system includes a human approval stage before finalizing the research report.

Workflow:

Generated Research Report



Human Review



Approved Final Report

This ensures users can verify AI-generated content before completion.

4. Advanced Features

Planning and Reasoning

The Planner Agent performs task decomposition by converting a broad research topic into smaller research activities.

Shared Memory

Implemented using centralized state management to preserve information between agent executions.

Reflection / Self Review

The Research Gap Agent analyzes existing findings and identifies missing research directions, improving the quality of generated reports.

Error Handling

The system includes:

- API configuration validation
- Exception handling during agent execution
- Output generation checks

Session Persistence

Generated research artifacts are stored locally, allowing users to access previous workflow outputs.

Future Extension Possibilities

Potential improvements include:

- Retrieval-Augmented Generation (RAG)
- Semantic Scholar/arXiv integration
- Persistent research history
- Web-based interface
- Citation verification system